

Practice Questions

Exercise 1

Consider the following dataset:

$$A = (0, 2), \quad B = (0, 1), \quad C = (1, 0).$$

The k -means algorithm is initialized with centers at A and B . Upon convergence, what will be the two centers of the clusters?

Solution:

The dataset is

$$A = (0, 2), \quad B = (0, 1), \quad C = (1, 0),$$

and the initial cluster centers are at

$$\mu_1^{(0)} = A = (0, 2), \quad \mu_2^{(0)} = B = (0, 1).$$

Step 1: Assign each point to the nearest center.

Using Euclidean distance:

$$d(A, \mu_1^{(0)}) = 0, \quad d(A, \mu_2^{(0)}) = 1$$

so A is assigned to cluster 1.

$$d(B, \mu_1^{(0)}) = 1, \quad d(B, \mu_2^{(0)}) = 0$$

so B is assigned to cluster 2.

For point $C = (1, 0)$:

$$d(C, \mu_1^{(0)}) = \sqrt{(1-0)^2 + (0-2)^2} = \sqrt{5},$$

$$d(C, \mu_2^{(0)}) = \sqrt{(1-0)^2 + (0-1)^2} = \sqrt{2}.$$

Since $\sqrt{2} < \sqrt{5}$, C is assigned to cluster 2.

Thus, after the first assignment:

$$\text{Cluster 1} = \{A\}, \quad \text{Cluster 2} = \{B, C\}.$$

Step 2: Update the cluster centers.

The new center of cluster 1 is

$$\mu_1^{(1)} = A = (0, 2).$$

The new center of cluster 2 is the mean of B and C :

$$\mu_2^{(1)} = \left(\frac{0+1}{2}, \frac{1+0}{2} \right) = \left(\frac{1}{2}, \frac{1}{2} \right).$$

Step 3: Reassign points using the updated centers.

For $A = (0, 2)$:

$$d(A, \mu_1^{(1)}) = 0,$$

$$d(A, \mu_2^{(1)}) = \sqrt{\left(0 - \frac{1}{2}\right)^2 + \left(2 - \frac{1}{2}\right)^2} = \sqrt{\frac{1}{4} + \frac{9}{4}} = \sqrt{\frac{10}{4}} = \sqrt{\frac{5}{2}}.$$

So A remains in cluster 1.

For $B = (0, 1)$:

$$d(B, \mu_1^{(1)}) = 1,$$

$$d(B, \mu_2^{(1)}) = \sqrt{\left(0 - \frac{1}{2}\right)^2 + \left(1 - \frac{1}{2}\right)^2} = \sqrt{\frac{1}{4} + \frac{1}{4}} = \sqrt{\frac{1}{2}}.$$

So B remains in cluster 2.

For $C = (1, 0)$:

$$d(C, \mu_1^{(1)}) = \sqrt{5},$$

$$d(C, \mu_2^{(1)}) = \sqrt{\left(1 - \frac{1}{2}\right)^2 + \left(0 - \frac{1}{2}\right)^2} = \sqrt{\frac{1}{4} + \frac{1}{4}} = \sqrt{\frac{1}{2}}.$$

So C remains in cluster 2.

The assignments do not change, so the algorithm has converged.

Final Answer:

Upon convergence, the two cluster centers are

$$\boxed{(0, 2) \quad \text{and} \quad \left(\frac{1}{2}, \frac{1}{2} \right)}.$$

Exercise 2

Consider the problem of binary classification using the Naive Bayes classifier. You are given two dimensional features (X_1, X_2) and the categorical class conditional distributions in the tables below. The entries in the tables correspond to $P(X_1 = x_1 \mid C_i)$ and $P(X_2 = x_2 \mid C_i)$ respectively. The two classes are *equally likely*. Given a data point $(1, -1)$, calculate $P(C_1 \mid X_1 = -1, X_2 = 1)$.

$X_1 =$ \ Class	C_1	C_2
-1	0.2	0.3
0	0.4	0.6
1	0.4	0.1

$X_2 =$ \ Class	C_1	C_2
-1	0.4	0.1
0	0.5	0.3
1	0.1	0.6

Figure 1:

Solution:

We are asked to compute

$$P(C_1 \mid X_1 = -1, X_2 = 1).$$

Since the two classes are equally likely,

$$P(C_1) = P(C_2) = \frac{1}{2}.$$

Using the Naive Bayes assumption,

$$P(X_1, X_2 \mid C_i) = P(X_1 \mid C_i) P(X_2 \mid C_i).$$

Therefore,

$$P(C_1 \mid X_1 = -1, X_2 = 1) = \frac{P(X_1 = -1 \mid C_1) P(X_2 = 1 \mid C_1) P(C_1)}{\sum_{i=1}^2 P(X_1 = -1 \mid C_i) P(X_2 = 1 \mid C_i) P(C_i)}.$$

From the tables:

$$P(X_1 = -1 \mid C_1) = 0.2, \quad P(X_2 = 1 \mid C_1) = 0.1,$$

$$P(X_1 = -1 \mid C_2) = 0.3, \quad P(X_2 = 1 \mid C_2) = 0.6.$$

Substituting these values:

$$P(C_1 \mid X_1 = -1, X_2 = 1) = \frac{(0.2)(0.1) \left(\frac{1}{2}\right)}{(0.2)(0.1) \left(\frac{1}{2}\right) + (0.3)(0.6) \left(\frac{1}{2}\right)}.$$

Since the factor $\frac{1}{2}$ appears in both numerator and denominator, it cancels:

$$P(C_1 \mid X_1 = -1, X_2 = 1) = \frac{(0.2)(0.1)}{(0.2)(0.1) + (0.3)(0.6)} = \frac{0.02}{0.02 + 0.18} = \frac{0.02}{0.20} = 0.1.$$

$$\boxed{P(C_1 \mid X_1 = -1, X_2 = 1) = 0.1}$$

Exercise 3

The table below (Table 1) is a distance matrix for 6 objects. Show the final result of hierarchical clustering with single link by drawing a dendrogram.

	<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>	<i>E</i>	<i>F</i>
<i>A</i>	0					
<i>B</i>	0.12	0				
<i>C</i>	0.51	0.25	0			
<i>D</i>	0.84	0.16	0.14	0		
<i>E</i>	0.28	0.77	0.70	0.45	0	
<i>F</i>	0.34	0.61	0.93	0.20	0.67	0

Table 1: Distance Table

Solution

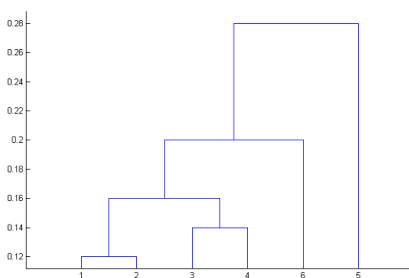


Figure 2: Solution for Exercise 3

Exercise 4

Show the final result of hierarchical clustering with complete link by drawing a dendrogram.

Solution

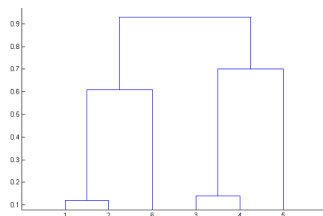


Figure 3: Solution for Exercise 4

Exercise 5

Change two values from the Table 1 so that your answer to the last two questions would be same.

Solution There is more than one way possible, but one way would be the following: The first step that the complete link clustering differs from the single link clustering is where AB and F are grouped together by $dist(AB, F) = dist(B, F) = 0.61$. We'd want $dist(AB, CD) = dist(A, D)$ to be smaller than this value, such as 0.53. Then we want $dist(ABCD, F) = dist(C, F) = 0.93$ to be the smallest so that ABCD and F are grouped together. We set this value to 0.63. After these changes both dendrograms become identical.

Exercise 6

Which clustering method(s) is most likely to produce the following results at $k = 2$? Choose the most likely method(s) and briefly explain why it/they will work better where others will not in **at most 3 sentences**. Here are the five clustering methods you can choose from:

- Hierarchical clustering with single link
- Hierarchical clustering with complete link
- Hierarchical clustering with average link
- k -means

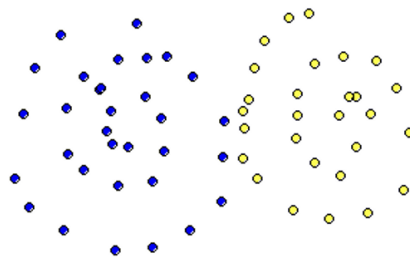


Figure 4: Data Distribution Within Clusters.

Solution K-means. Hierarchical clustering wouldn't work since the early few steps will group instances near the decision boundary (note some of them are very close).

Exercise 7

Suppose that an IR system contains only 1000 documents numbered d_1 to d_{1000} . A query is known to generate 27 relevant documents as listed below:

$$\{d_1, d_5, d_7, d_{10}, d_{88}, d_{151}, d_{200}, d_{211}, d_{250}, d_{300}, d_{399}, d_{401}, d_{405}, \\ d_{450}, d_{473}, d_{500}, d_{501}, d_{530}, d_{545}, d_{590}, d_{600}, d_{735}, d_{700}, d_{720}, d_{800}, d_{888}, d_{900}\}.$$

Two very different IR systems are used to retrieve ranked documents for this query. Each system only returns the top 10 ranked documents in the following order of ranking. Systems 1 and 2 each retrieve documents one at a time in the following order with all 10 documents eventually returned:

- **System 1:** $d_{122}, d_{211}, d_{150}, d_{88}, d_{37}, d_1, d_{501}, d_{800}, d_{201}, d_5$.
- **System 2:** $d_{10}, d_6, d_{700}, d_{250}, d_{88}, d_{600}, d_{59}, d_{422}, d_{500}, d_{333}$.

Answer the following and show your work:

- Calculate the F1 score for each.
- Which IR system is better? Justify your answer.

Solution:

The F1 score is defined as

$$F_1 = \frac{2PR}{P + R},$$

where P is precision and R is recall.

There are 27 relevant documents in total.

System 1:

Retrieved documents:

$$d_{122}, d_{211}, d_{150}, d_{88}, d_{37}, d_1, d_{501}, d_{800}, d_{201}, d_5$$

Among these, the relevant ones are:

$$d_{211}, d_{88}, d_1, d_{501}, d_{800}, d_5$$

So System 1 retrieves 6 relevant documents out of 10.

$$P_1 = \frac{6}{10} = 0.6$$

$$R_1 = \frac{6}{27} = \frac{2}{9}$$

Thus,

$$F_{1,1} = \frac{2 \cdot 0.6 \cdot \frac{2}{9}}{0.6 + \frac{2}{9}}$$

Writing $0.6 = \frac{3}{5}$,

$$F_{1,1} = \frac{2 \cdot \frac{3}{5} \cdot \frac{2}{9}}{\frac{3}{5} + \frac{2}{9}} = \frac{\frac{4}{15}}{\frac{27+10}{45}} = \frac{\frac{4}{15}}{\frac{37}{45}} = \frac{4}{15} \cdot \frac{45}{37} = \frac{12}{37} \approx 0.324$$

System 2:

Retrieved documents:

$$d_{10}, d_6, d_{700}, d_{250}, d_{88}, d_{600}, d_{59}, d_{422}, d_{500}, d_{333}$$

Among these, the relevant ones are:

$$d_{10}, d_{700}, d_{250}, d_{88}, d_{600}, d_{500}$$

So System 2 also retrieves 6 relevant documents out of 10.

$$P_2 = \frac{6}{10} = 0.6$$

$$R_2 = \frac{6}{27} = \frac{2}{9}$$

Thus,

$$F_{1,2} = \frac{2 \cdot 0.6 \cdot \frac{2}{9}}{0.6 + \frac{2}{9}} = \frac{12}{37} \approx 0.324$$

Final Answer:

$$F_1(\text{System 1}) = \frac{12}{37} \approx 0.324$$

$$F_1(\text{System 2}) = \frac{12}{37} \approx 0.324$$

Therefore, both systems have the same F1 score.

Exercise 8

An IR system returns 8 relevant documents, and 10 nonrelevant documents. There are a total of 20 relevant documents in the collection. What is the precision of the system on this search, and what is its recall?

Solution: The system returns:

- Relevant documents retrieved = 8
- Nonrelevant documents retrieved = 10

Total documents retrieved:

$$8 + 10 = 18$$

Total relevant documents in the collection:

$$20$$

Precision:

$$\text{Precision} = \frac{\text{Relevant retrieved}}{\text{Total retrieved}} = \frac{8}{18} = \frac{4}{9} \approx 0.444$$

Recall:

$$\text{Recall} = \frac{\text{Relevant retrieved}}{\text{Total relevant}} = \frac{8}{20} = 0.4$$

Final Answer:

$$\text{Precision} = \frac{4}{9} \approx 0.444, \quad \text{Recall} = 0.4$$

Exercise 9

Consider an information need for which there are 4 relevant documents in the collection. Contrast two systems run on this collection. Their top 10 results are judged for relevance as follows (the leftmost item is the top ranked search result):

- **System 1:** R N R N N N N N R R
- **System 2:** N R N N R R R N N N

1. What is the AP of each system? Which has a higher AP?
2. What is the R-precision of each system? (Does it rank the systems the same as AP?)

Solution:

System 1:

Ranking: *R N R N N N N R R*

The relevant documents occur at ranks 1, 3, 9, 10.

$$P@1 = \frac{1}{1}$$

$$P@3 = \frac{2}{3}$$

$$P@9 = \frac{3}{9} = \frac{1}{3}$$

$$P@10 = \frac{4}{10} = \frac{2}{5}$$

Thus,

$$AP_1 = \frac{1}{4} \left(1 + \frac{2}{3} + \frac{1}{3} + \frac{2}{5} \right)$$

$$AP_1 = \frac{1}{4} \left(2 + \frac{2}{5} \right) = \frac{1}{4} \cdot \frac{12}{5} = \frac{3}{5} = 0.6$$

System 2:

Ranking: *N R N N R R R N N N*

The relevant documents occur at ranks 2, 5, 6, 7.

$$P@2 = \frac{1}{2}$$

$$P@5 = \frac{2}{5}$$

$$P@6 = \frac{3}{6} = \frac{1}{2}$$

$$P@7 = \frac{4}{7}$$

Thus,

$$AP_2 = \frac{1}{4} \left(\frac{1}{2} + \frac{2}{5} + \frac{1}{2} + \frac{4}{7} \right)$$

$$AP_2 = \frac{1}{4} \left(1 + \frac{2}{5} + \frac{4}{7} \right) = \frac{1}{4} \left(1 + \frac{14}{35} + \frac{20}{35} \right) = \frac{1}{4} \left(1 + \frac{34}{35} \right)$$

$$AP_2 = \frac{1}{4} \cdot \frac{69}{35} = \frac{69}{140} \approx 0.493$$

$$AP(\text{System 1}) = 0.6$$

$$AP(\text{System 2}) = \frac{69}{140} \approx 0.493$$

Answer: System 1 has the higher AP.

R-Precision Solution: R-precision is defined as the precision at rank R , where R is the total number of relevant documents. Here, $R = 4$.

System 1:

Top 4 results: $R \ N \ R \ N$

Number of relevant documents in top 4 = 2

$$R\text{-precision}_1 = \frac{2}{4} = 0.5$$

System 2:

Top 4 results: $N \ R \ N \ N$

Number of relevant documents in top 4 = 1

$$R\text{-precision}_2 = \frac{1}{4} = 0.25$$

Conclusion:

$$R\text{-precision}_1 = 0.5 \quad > \quad R\text{-precision}_2 = 0.25$$

Thus, System 1 also ranks higher than System 2 under R-precision. This is consistent with the AP ranking.

Exercise 10

Which of the following expressions is equivalent to $p(A \mid B, C, D)$?

(a)

$$\frac{p(A, B, C, D)}{p(C \mid B, D) p(B \mid D) p(D)}$$

(b)

$$\frac{p(A, B, C, D)}{p(B, C) p(D)}$$

(c)

$$\frac{p(A, B, C, D)}{p(B, C \mid D) p(B) p(C)}$$

Solution Answer is (a).

$$p(A \mid B, C, D) = \frac{p(A, B, C, D)}{p(B, C, D)} = \frac{p(A, B, C, D)}{p(C \mid B, D) p(B, D)} = \frac{p(A, B, C, D)}{p(C \mid B, D) p(B \mid D) p(D)}.$$